

IMPORT DATA USING TRANSFORM MAPS

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In partial fulfillment for the award of the degree of

BACHELOR OF SCIENCE IN ARTIFICIAL INTELLIGENCE



DEPARTMENT OF ARTIFICIAL INTELLIGENCE

MARIA ARTS AND SCIENCE COLLEGE FOR WOMEN

(AFFILIATED TO MANONMANIAM SUNDARANAR

UNIVERSITY, TIRUNELVELI)

VALLIOOR – 627117

2026-2027

IMPORT DATA USING TRANSFORM MAPS

Project Overview

The project titled “**Import Data Using Transform Maps**” is developed using the ServiceNow platform to demonstrate the process of importing employee data from an external source into a target ServiceNow table.

The project uses **Import Sets and Transform Maps** to transfer, map, and transform employee information. The source data is first imported into ServiceNow and then mapped to the appropriate target fields using a Transform Map.

The main purpose of this project is to understand the complete data-import process and verify that the employee records are correctly transferred into the target table.

Objectives

- To import employee data into ServiceNow.
- To understand the Import Set process.
- To create and configure Transform Maps.
- To map source fields with target fields.
- To reduce manual data entry.
- To verify the transformed employee records.
- To understand the complete data migration workflow.

2. IDEATION PHASE

2.1 Brainstorming

The project idea was developed by identifying the difficulties involved in manually entering employee information into ServiceNow.

Manual data entry for a large number of records can take considerable time and may result in incorrect or inconsistent information. Therefore, different approaches for transferring employee data were discussed.

The selected approach was to use **ServiceNow Import Sets and Transform Maps**. Import Sets allow the source data to be brought into ServiceNow, while Transform Maps define how the imported fields should be transferred to the target table.

The brainstorming process resulted in the project idea of demonstrating employee data import using Transform Maps.

2.2 Define Problem

Employee information may be maintained in external files such as CSV or Excel. Entering every record manually into ServiceNow can be time-consuming.

The major problems include:

- Manual data entry.
- Incorrect data entry.
- Incorrect field mapping.
- Missing employee information.
- Duplicate records.
- Difficulty in handling large amounts of data.

The project addresses these issues by providing a structured process for importing employee data and mapping it to the required target fields using Transform Maps.

2.3 Empathy Map

Think and Feel

The user expects the data-import process to be simple, accurate, and easy to verify.

See

The user sees the employee source file, Import Set records, Transform Map, field mappings, transformation results, and final employee records.

Say and Do

The user prepares the employee data, uploads the file, configures the Transform Map, executes the transformation, and verifies the final records.

Pains

- Manual data entry
- Field mapping errors
- Missing information
- Duplicate records
- Data-format problems

Gains

- Reduced manual work
- Faster data import
- Organized data migration
- Easy verification
- Accurate field mapping

3. REQUIREMENT ANALYSIS

3.1 Customer Journey

The customer journey consists of the following steps:

Prepare Employee Data → Upload Data → Import Set → Verify Source Records → Create Transform Map → Configure Field Mapping → Execute Transform → Verify Target Records

The user first prepares the employee data in a source file. The file is then uploaded into ServiceNow using Import Sets.

After the source records are imported, a Transform Map is created. The source fields are mapped to the corresponding target fields.

Finally, the transformation is executed and the resulting employee records are verified in the target table.

3.2 Data Flow Diagram

The data flow of the project is:

Employee Source File



Import Set



Import Set Table



Transform Map



Field Mapping



Target Employee Table



Final Employee Records

The source file contains employee information. The data is imported into the Import Set table. The Transform Map processes the source data and transfers it to the appropriate target fields.

3.3 Solution Requirement

The project requires the following:

- ServiceNow instance
- Employee source data
- CSV or Excel file
- Import Set
- Import Set table
- Target table
- Transform Map
- Field Maps
- Web browser
- Internet connection

The system should allow the user to upload employee data, configure field mappings, execute the transformation, and verify the final records.

3.4 Technology Stack

Platform: ServiceNow

Features Used:

- Import Sets
- Import Set Tables
- Transform Maps
- Field Maps
- Target Tables
- Import/Transform Results

Data Source: CSV / Excel

Browser: Google Chrome

Environment: ServiceNow Developer/Learning Instance

4. PROJECT DESIGN

The project is designed using three major solutions.

4.1 Solution 1 – Source Data Preparation

The first solution focuses on preparing the employee source data.

The source file contains employee information such as:

- Employee ID
- Name
- Email
- Department
- Location

The data is checked before importing to ensure that the required fields contain valid information.

4.2 Solution 2 – Import Set and Transform Map

The second solution focuses on importing the source data into ServiceNow.

The employee file is uploaded using Import Sets. The imported information is stored in the Import Set table.

A Transform Map is then created to connect the source fields with the corresponding fields in the target table.

Example:

Source Field	Target Field
Employee ID	Employee ID
Name	Name
Email	Email
Department	Department
Location	Location

4.3 Solution 3 – Validation and Verification

The third solution focuses on verifying the transformed data.

After executing the Transform Map, the target table is checked to ensure that the employee records have been transferred correctly.

The source and target values are compared, and the import/transform results are checked for errors.

5. PROJECT PLANNING

The project is planned in the following sequence:

1. Understand the project requirement.
2. Prepare employee source data.
3. Create or identify the Import Set.
4. Upload the source file.
5. Verify imported records.
6. Identify the target table.
7. Create the Transform Map.
8. Configure field mappings.
9. Execute the Transform.
10. Verify the target records.
11. Perform testing.
12. Perform User Acceptance Testing.

13. Capture screenshots.
14. Prepare project documentation.
15. Record the demonstration video.

The project planning ensures that each stage is completed and verified systematically.

6. PROJECT DEVELOPMENT

6.1 Source Data Preparation

The employee data is prepared in a structured source file. The required fields are added with appropriate values.

6.2 Importing Data

The source file is uploaded into ServiceNow using the Import Set functionality.

The imported data is stored temporarily in the Import Set table.

6.3 Creating Transform Map

A Transform Map is created by selecting the required target table.

The source fields are mapped to their corresponding target fields.

6.4 Field Mapping

Field Maps are configured to define the relationship between source and target fields.

For example:

Employee ID → Employee ID

Name → Name

Email → Email

Department → Department

Location → Location

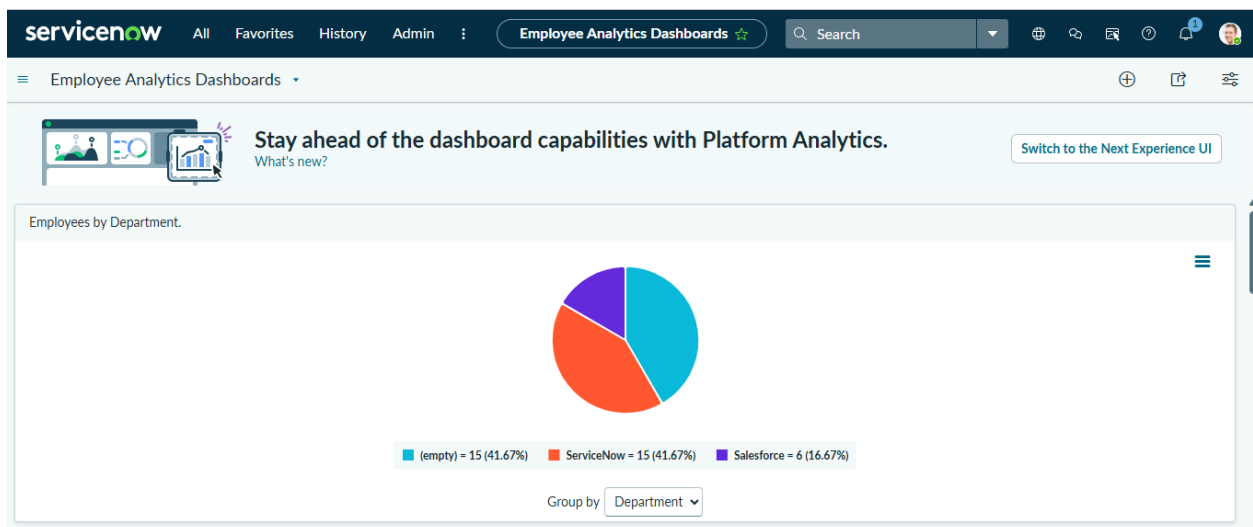
6.5 Executing Transform

After completing the field mappings, the Transform Map is executed.

The source data is processed and transferred into the target table.

6.6 Verification

The target table is opened and the employee records are checked to ensure that the data has been transferred correctly.



7. TESTING

Testing is performed to verify whether the Import Data Using Transform Maps project works correctly.

Test Case 1 – Source File Upload

Test: Upload the employee source file.

Expected Result: The source data should be successfully imported into the Import Set.

Test Case 2 – Import Set Verification

Test: Check the imported source records.

Expected Result: Employee records should be displayed correctly in the Import Set table.

Test Case 3 – Field Mapping

Test: Verify the source-to-target field mappings.

Expected Result: Each source field should be mapped to the appropriate target field.

Test Case 4 – Transform Execution

Test: Execute the Transform Map.

Expected Result: Employee records should be created or updated in the target table.

Test Case 5 – Target Table Verification

Test: Check the final employee records.

Expected Result: Employee information should appear correctly in the target table.

8. USER ACCEPTANCE TESTING

User Acceptance Testing is performed to verify that the project satisfies the expected requirements.

UAT-01 – Upload Data

Action: Upload employee source data.

Expected Result: Data should be successfully imported.

UAT-02 – Verify Source Records

Action: Open the Import Set records.

Expected Result: Imported employee records should be visible.

UAT-03 – Verify Field Mapping

Action: Check the Transform Map field mappings.

Expected Result: Source fields should be connected to the correct target fields.

UAT-04 – Execute Transform

Action: Run the Transform Map.

Expected Result: Employee records should be transferred to the target table.

UAT-05 – Verify Final Records

Action: Open the target employee table.

Expected Result: The imported employee records should be displayed correctly.

UAT-06 – Verify Errors

Action: Check import and transform results.

Expected Result: Any failed or skipped records should be identifiable for correction.

9. RESULTS AND DISCUSSION

The project successfully demonstrates the process of importing employee data into ServiceNow using Import Sets and Transform Maps.

The source employee data is uploaded into ServiceNow through the Import Set functionality. The data is then processed using the configured Transform Map.

The source fields are mapped to their corresponding target fields, allowing the employee information to be transferred into the target table.

After transformation, the target records are verified to ensure that the imported information is displayed correctly.

The project provides practical understanding of:

- Import Set creation
- Source data importing
- Transform Map configuration
- Field mapping
- Data transformation
- Target record verification
- Import and transformation results

[Add your actual result screenshot here]

10. CONCLUSION

The **Import Data Using Transform Maps** project demonstrates a structured method of transferring employee information into ServiceNow.

The project explains the complete workflow from preparing the source data to importing the records, configuring the Transform Map, mapping the fields, executing the transformation, and verifying the final records.

Through this project, the working process of ServiceNow Import Sets and Transform Maps is understood practically.

The project also demonstrates how a systematic data-import process can reduce repetitive manual data entry and provide an organized method for transferring information between a source file and a ServiceNow target table.

11. FUTURE ENHANCEMENT

The project can be enhanced with additional features in the future.

1. Automated Data Import

The employee data import process can be scheduled to run automatically at regular intervals.

2. Advanced Data Validation

Additional validation rules can be implemented to identify invalid email addresses, missing fields, incorrect formats, and duplicate records.

3. Error Handling

Detailed error-handling mechanisms can be added to identify and report failed records automatically.

4. Automated Notifications

Notifications can be configured to inform administrators about successful or failed import operations.

5. Scheduled Imports

Scheduled data imports can be implemented for regularly updated employee information.

6. Data Transformation Scripts

Transform Scripts can be used when advanced data-processing logic is required.

7. Import Monitoring

Import results can be monitored through dashboards or reports to provide better visibility into the data-import process.

12. REFERENCES

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2. ServiceNow Developer Documentation – Data Import and Transformation.
3. ServiceNow Learning Resources – Import Sets and Transform Maps.
4. ServiceNow Developer/Learning Instance – Practical Implementation.
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